

HXC-108_ios_evidence

HXC-108 — HelixCode iOS App Fully-Automatic Simulator Recording Evidence

Revision	1
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Status	active
Status summary	PASS — app built, launched, recorded, OCR-validated on iOS Simulator
Feature class	§11.4.69 <code>video_display</code> (UI render) + §11.4.158 intensive recording coverage

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Summary

The HelixCode iOS app (`helix_code/applications/ios/HelixCode.xcodeproj`, scheme `HelixCode`) was built, installed, launched, and screen-recorded **fully automatically** on the iOS Simulator — no TCC grant, no human interaction. The recording and screenshots show the **real** app UI rendering **live data from the embedded Go core** (`HelixCore.xcframework` over `cgo`): Go core OK — themes: 3, tasks: 2 plus two tasks seeded through the core. Content was verified by macOS Vision OCR, and the OCR analyzer was self-validated (golden-good accepted, golden-bad black frame rejected) per §11.4.107(10).

This was NOT a clean first build — a real CoreSimulator/environment blocker had to be root-caused and surmounted with a reversible host-side fix (documented below). No step was faked.

Environment (captured)

- Xcode 16.4 (Build 16F6); iOS Simulator SDK 18.5.
- Simulator: **iPhone 16, iOS 18.5** (UDID `DC8B1905-F2E5-425D-A2DE-42C27B390AC5`).

- Host: arm64 (Apple Silicon). `HelixCore.xcframework` ships an `ios-arm64_x86_64-simulator` slice (arch `x86_64 arm64`) — compatible.
- Project: single scheme `HelixCode`; `CODE_SIGNING_ALLOWED=NO`; bundle id `dev.helix.code.ios`; embeds `HelixCore.xcframework` (the real Go mobile core).

Buildability

PASS. Build command (final, succeeding):

```
cd helix_code/applications/ios && \
xcodebuild -project HelixCode.xcodeproj -scheme HelixCode -configuration D
  -destination 'platform=iOS Simulator,id=DC8B1905-F2E5-425D-A2DE-42C27B39
  -derivedDataPath /tmp/helixcode_ios_dd CODE_SIGNING_ALLOWED=NO build
# => ** BUILD SUCCEEDED ** (EXIT=0)
```

Produced `.app` (real, well-formed):

- `HelixCode` — Mach-O 64-bit executable arm64 (simulator slice).
- `Frameworks/HelixCore.framework` embedded.
- `HelixCode.debug.dylib` 51.5 MB (the substantial Go core — not a stub).
- `Assets.car` (284 KB), `Info.plist` (`CFBundleIdentifier=dev.helix.code.ios`).

Swift compilation, the gomobile Go-core link, and `___preview.dylib` all succeeded on the very first attempt; only the asset-catalog stage failed initially (see below).

Build-blocker root cause and reversible fix

Initial failure (EXIT=65): `actool` (asset-catalog `CompileAssetCatalogVariant` thinned) could not create its transient `IBSimDeviceTypeiPad3x` render device for iOS 18.5 — “Device was allocated but was stuck in creation state.”

Root cause (FACT, from `~/Library/Logs/CoreSimulator/CoreSimulator.log`):

```
Error copying sample content to path
/Volumes/T7/Library-Developer-Xcode/UserData/IB Support/Simulator Device
NSCocoaErrorDomain Code=513 "You don't have permission to save the file
NSUnderlyingError ... NSPOSIXErrorDomain Code=1 "Operation not permitted
```

- `~/Library/Developer/Xcode` is a symlink to `/Volumes/T7/Library-Developer-Xcode`, so the IB device set physically lives on the **T7 external volume**, mounted `hfs, local, noowners` (`mount | grep /Volumes/T7`).
- Under `noowners`, the sandboxed `CoreSimulatorService` daemon cannot create the device’s data dir (Operation not permitted), even though the shell user can write there. A normal `simctl create` into the *default* device set (`~/Library/Developer/CoreSimulator/Devices`, internal disk) succeeds — proving the failure is specific to the relocated IB set on the `noowners` volume.
- The project’s `TARGETED_DEVICE_FAMILY="1,2"` forces `actool` to need an iPad-3x render device; `TARGETED_DEVICE_FAMILY=1`, `IBSIMULATOR_DEVICE_SET_PATH`, and `ASSETCATALOG_COMPILER_SKIP_APP_STORE_DEPLOYMENT=YES` overrides were all tried and did **not** bypass it (the path is baked into `actool`’s default device-set).

Reversible fix applied: relocated the IB Simulator Devices directory off the noowners T7 volume onto an internal owners-enforced path (/System/Volumes/Data/..., daemon-writable) via a symlink, preserving the original as Simulator Devices.t7bak. Rebuild then **succeeded**. Fix fully reverted afterward (symlink removed, original restored — see Cleanup). This is a **host-environment** blocker, not a HelixCode source defect.

Recording + OCR validation

- Recording: helixcode-ios-launch-render-20260623-141821.mp4
 - /Volumes/T7/Downloads/Recordings/ (raw corpus, gitignored per §11.4.128).
 - md5 e565c6bde216926cbb3d21990bfd6c5d; H.264; 1178x2556; **144 frames; 11.945 s**; 3.17 MB (via simctl io recordVideo — window-scoped to the simulator device, no desktop, §11.4.159(A)).
 - §11.4.155 project-name prefix helixcode-ios- (from lowercased project root).
- Curated still: helixcode-ios-launch-render-20260623-141821.png.

Liveness (§11.4.107): per-frame avg luma varies across the timeline (44 → 118 light-appearance toggle → 78 → ~16 relaunch transition → back to 44); 6+ distinct frame hashes — genuine motion, NOT a frozen/black frame.

Content (macOS Vision OCR — robust on dark UI where tesseract returned empty): 9 of 12 sampled recording frames (f_001–f_008, f_012) contain the full app UI; f_009–f_011 are the relaunch black-launch-screen transition (matching the luma dip). OCR of an app frame / screenshot yields, verbatim:

```
HelixCode
HelixCode iOS
Disconnected
User: (none)
Go core OK - themes: 3, tasks: 2
Connect
Build iOS client - created
Wire Go core - created
```

These are §11.4.2 real outputs — Go core OK — themes: 3, tasks: 2 and the two task rows are **live data from the Go core** (getAvailableThemes(), getTasks(), createTask()), not hardcoded/simulated. Disconnected is correct (no server runs in the simulator). No “simulated”/“placeholder”/“TODO” text.

All §11.4.159(K) expected patterns present: HelixCode iOS, Disconnected, Go core OK - themes: 3, tasks: 2, Connect, Build iOS client, Wire Go core.

Durable evidence (committed) — rotation-proof anchors (§11.4.83, HXC-108 audit F2 fix)

The raw corpus (/Volumes/T7/Downloads/Recordings/, §11.4.128/154-rotatable) is the secondary location. The load-bearing key still frame is **copied into the committed tree**; the 3.17 MB MP4 exceeds the per-byte commit budget, so its identity is pinned by a committed ffp+md5 stamp (the still frame is the durable visual evidence):

Committed artifact	sha256
docs/qa/HXC-108_ios/helixcode-ios-launch-render-20260623-141821.png	4467408ec80b9097a7f7461915f32046a9eb2
docs/qa/HXC-108_ios/helixcode-ios-launch-render-20260623-141821.mp4.ffprobe-stamp.txt (pins the raw MP4: h264 1178×2556 144f 11.945s, md5 e565c6bde216926cbb3d21990bfd6c5d)	(text stamp)

Analyzer self-validation

Per §11.4.107(10) — the OCR analyzer provably cannot bluff:

- golden-bad (synthetic black frame) → HelixCode iOS matches = **0** (rejected).
- golden-good (app screenshot) → HelixCode iOS matches = **1** (accepted).
- Verdict: **ANALYZER SELF-VALIDATION PASS.**

Cleanup / reversibility

- App terminated; simulator DC8B1905 - . . . powered off — restored to its pre-run Shutdown state (§11.4.14; we booted it, so we shut it down). Other simulators untouched.
- IB Simulator Devices relocation fully reverted: symlink removed, original T7 directory restored from .t7bak.
- Raw recording corpus under /Volumes/T7/Downloads/Recordings/ is gitignored per §11.4.128; this evidence doc is the curated committable record (§11.4.83).
- No commit/push performed — returned for conductor review (§11.4.142).

Verdict

PASS — fully-automatic iOS Simulator build + launch + recording achieved with captured, OCR-validated, self-validated evidence of the real HelixCode iOS app rendering live Go-core data. The only obstacle was a host-environment CoreSimulator permission blocker (noowners external volume), root-caused and surmounted reversibly — not a code defect, not faked.